Q1. Define Blockchain Technology. Explain its importance as a Distributed Ledger Technology (DLT).

Definition

Blockchain is a decentralized and distributed digital ledger technology that records transactions in a series of interconnected blocks. Once information is recorded and validated, it is very difficult to alter it without detection.

Each block generally contains:

- Transaction/data information
- Timestamp
- Hash of the current block
- Hash of the previous block

Blockchain as Distributed Ledger Technology

A **Distributed Ledger Technology (DLT)** is a system in which transaction records are stored and maintained across multiple computers or nodes instead of being controlled by a single central authority.

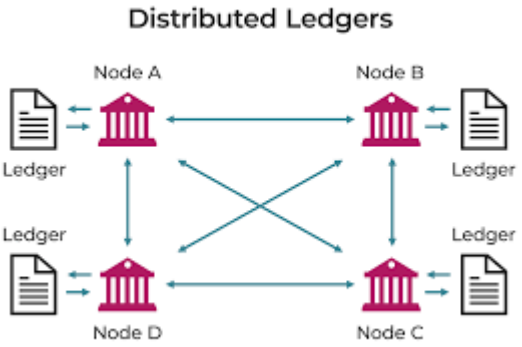
Blockchain is an important type of DLT because:

1. The ledger is distributed among multiple participants.
2. Transactions are verified using a consensus mechanism.
3. Cryptography protects the integrity of records.
4. Each block is linked to the previous block using hashes.
5. Participants can maintain a synchronized copy of the ledger.

Importance

- Provides transparency.
- Reduces dependence on a central authority.
- Makes unauthorized modification difficult.
- Improves trust between participants.
- Provides traceable and auditable records.

Example: Bitcoin uses blockchain to maintain a distributed record of cryptocurrency transactions.



Q2. Explain the Main Characteristics/Features of Blockchain Technology.

The major characteristics of blockchain are:

1. Decentralization

Blockchain distributes control and data among multiple nodes instead of depending on a single central authority.

2. Immutability

Once a transaction is confirmed and added to the blockchain, changing it is extremely difficult because blocks are cryptographically linked.

3. Transparency

In many blockchain networks, participants can view and verify transaction records according to the network's access rules.

4. Security

Cryptographic techniques such as hashing and digital signatures help protect transactions and records.

5. Distributed Ledger

Copies of the ledger are maintained across multiple participating nodes.

6. Consensus

Network participants use a consensus mechanism to agree on which transactions should be accepted.

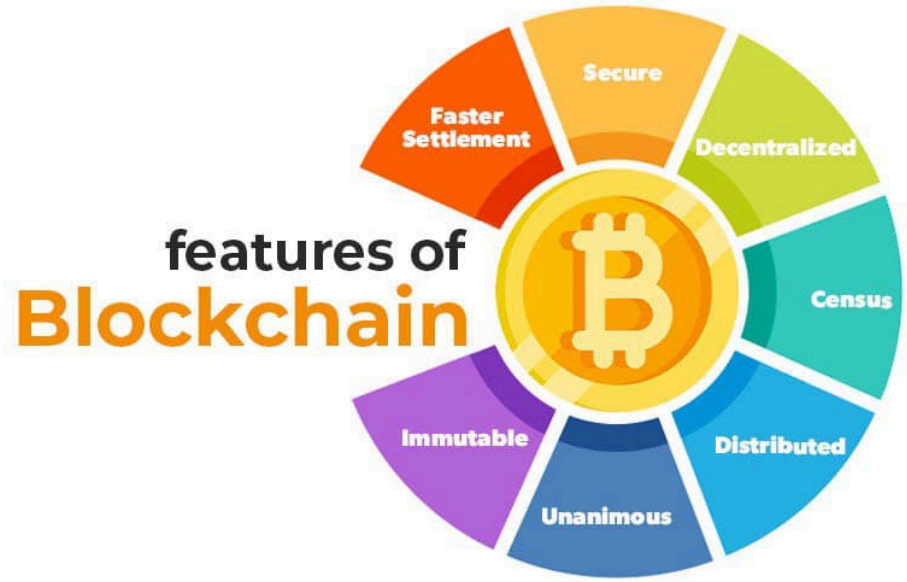
7. Traceability

Transactions can be tracked through the chain of records, providing an auditable history.

8. Programmability

Some blockchain platforms, such as Ethereum, support smart contracts that automatically execute programmed rules.

Conclusion: These features make blockchain useful for applications requiring trusted, verifiable, and tamper-resistant records.



Q3. Explain the Three Important Features of Blockchain: Decentralization, Immutability, and Enhanced Security.

1. Decentralization

Decentralization means that blockchain does not necessarily depend on a single central authority. Multiple nodes participate in maintaining and validating the ledger.

Benefits:

- Reduces dependence on a single point of control.
- Improves availability.

- Makes some forms of centralized manipulation more difficult.

Example: Bitcoin transactions are validated by participating nodes rather than by one central bank.

2. Immutability

Immutability means that confirmed blockchain records are designed to be highly resistant to unauthorized modification.

Each block contains a cryptographic hash that connects it to the previous block. Changing historical data would therefore affect the chain's cryptographic relationships and may be detected or rejected by the network.

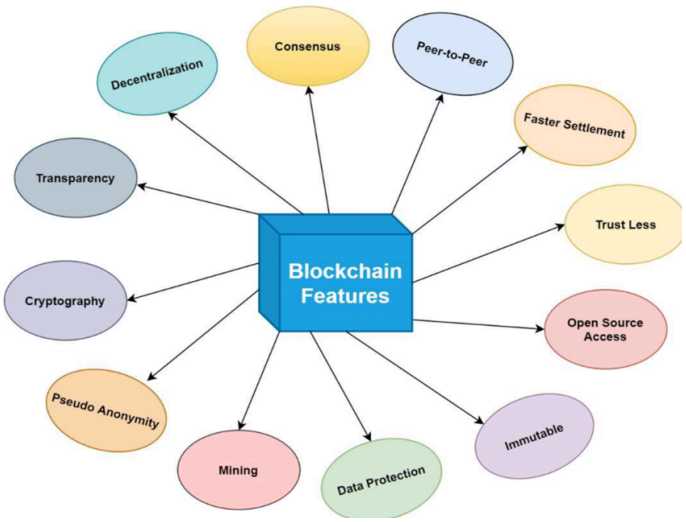
3. Enhanced Security

Blockchain uses several security mechanisms, including:

- Cryptographic hashing
- Digital signatures
- Access controls
- Consensus mechanisms
- Distributed storage

These mechanisms make unauthorized alteration and fraudulent transactions more difficult.

Conclusion: Decentralization, immutability, and cryptographic security together form important foundations of blockchain technology.



Q4. Explain the Steps Involved in the Working of Blockchain Technology.

The basic working of a blockchain can be explained in the following steps:

Step 1: Transaction Creation

A user creates and digitally signs a transaction.

Step 2: Transaction Broadcasting

The transaction is broadcast to participating nodes in the blockchain network.

Step 3: Transaction Verification

Network nodes verify the transaction according to the blockchain's rules.

Step 4: Consensus

The network uses a consensus mechanism to determine whether the transaction should be accepted.

Step 5: Block Formation

Valid transactions are grouped together into a block.

Step 6: Block Addition

The accepted block is added to the existing blockchain.

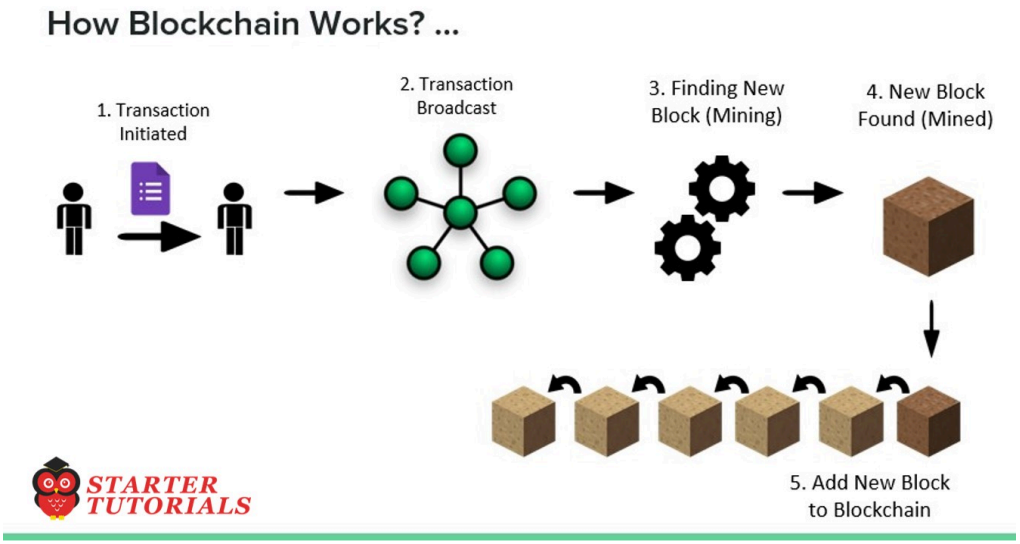
Step 7: Ledger Update

Participating nodes update their copies of the ledger according to the accepted blockchain state.

Simple Flow

Transaction → Broadcast → Verification → Consensus → Block Creation → Block Added → Ledger Updated

Conclusion: Blockchain provides a systematic process for recording and validating transactions without necessarily requiring a central authority.



Q5. What is Consensus in Blockchain? Explain Its Role in Validating Transactions.

Meaning of Consensus

Consensus is the process through which participants in a blockchain network agree on the valid state of the distributed ledger.

Since blockchain networks can have many independent nodes, a mechanism is needed to determine which transactions and blocks should be accepted.

Role of Consensus

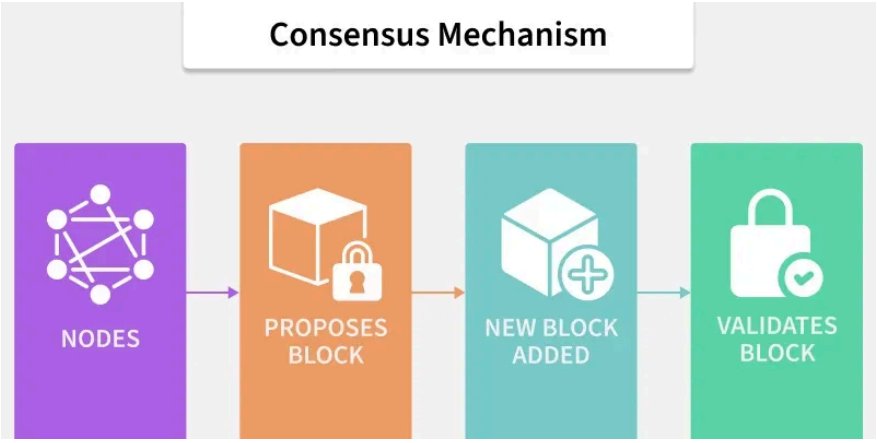
1. **Validates Transactions**
   It helps ensure that transactions follow the network's rules.
2. **Prevents Invalid Records**
   Invalid or conflicting transactions can be rejected.
3. **Maintains Ledger Consistency**
   Participating nodes can agree on the accepted blockchain state.
4. **Reduces Double Spending**
   In cryptocurrency systems, consensus helps determine which conflicting transaction should be accepted.
5. **Supports Decentralization**
   Agreement can be achieved without relying entirely on a single central authority.

Examples of Consensus Mechanisms

- **Proof of Work (PoW)**

- **Proof of Stake (PoS)**
- **Practical Byzantine Fault Tolerance (PBFT)**

Conclusion: Consensus is essential for maintaining trust, consistency, and validity in a distributed blockchain network.



Q6. Explain Public Blockchain and Private Blockchain with Suitable Examples.

Public Blockchain

A **public blockchain** is generally open to anyone who wants to participate according to the network's rules.

Features:

- Open participation.
- High level of decentralization.
- Transactions may be publicly verifiable.
- Uses a consensus mechanism.
- No single organization necessarily controls the network.

Examples: Bitcoin and Ethereum.

Private Blockchain

A **private blockchain** is controlled by an organization or authorized group. Participation is restricted to approved users.

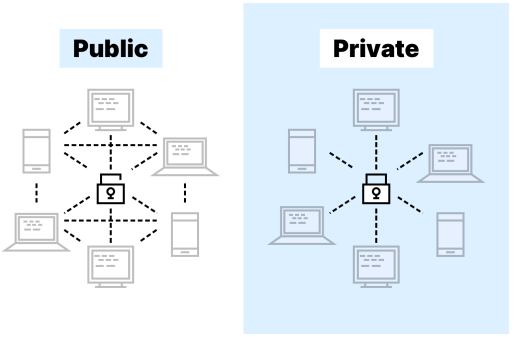
Features:

- Permission-based access.
- Controlled participation.
- Greater organizational control.
- Suitable for enterprise applications.
- Can provide faster processing in some configurations.

Examples: Hyperledger Fabric deployments and enterprise blockchain networks.

Difference

Public Blockchain	Private Blockchain
Open participation	Restricted participation
Generally more decentralized	Controlled by an organization/group
Anyone may participate according to network rules	Only authorized participants can participate
Suitable for open networks	Suitable for enterprise/internal use
Example: Bitcoin	Example: Hyperledger Fabric



Q7. Explain Any Two Blockchain Protocols: Hyperledger Fabric, Ethereum, Corda, and Quorum.

1. Hyperledger Fabric

Hyperledger Fabric is an enterprise-focused, permissioned blockchain platform.

Features:

- Permissioned network.
- Known participants.
- Modular architecture.
- Supports smart contracts, called **chaincode**.
- Suitable for business applications.

- Supports channels for communication and data isolation.

Applications: Supply-chain management, finance, healthcare, and enterprise record sharing.

2. Ethereum

Ethereum is a blockchain platform designed to support decentralized applications and smart contracts.

Features:

- Supports smart contracts.
- Supports decentralized applications (DApps).
- Uses the Ethereum Virtual Machine (EVM).
- Has a native cryptocurrency called **Ether (ETH)**.
- Can support many types of decentralized applications.

Applications: DeFi, tokenization, NFTs, decentralized applications, and automated agreements.

Q8. Explain the Role of Cryptography, Decentralization, and Consensus in Blockchain Security.

1. Cryptography

Cryptography protects blockchain data and transactions.

Important techniques include:

- **Hashing:** Creates a fixed-size representation of data and helps detect changes.
- **Digital signatures:** Help verify the authenticity of transactions.
- **Public and private keys:** Enable users to securely authorize transactions.

2. Decentralization

Instead of storing and controlling the ledger at one central location, blockchain can distribute it among multiple nodes.

This:

- Reduces dependence on a single point of failure.
- Makes unauthorized centralized modification more difficult.
- Improves resilience.

3. Consensus

Consensus allows participating nodes to agree on valid transactions and the accepted state of the blockchain.

It:

- Helps reject invalid transactions.
- Maintains consistency among nodes.
- Helps prevent certain forms of fraud and conflicting records.

Conclusion

Cryptography protects the data, decentralization distributes control, and consensus establishes agreement. Together, they provide important security foundations for blockchain systems.

Q9. Differentiate Between Blockchain and Bitcoin Based on Definition, Scope, Purpose, and Flexibility.

Basis	Blockchain	Bitcoin
Definition	A distributed ledger technology used to record data and transactions.	A decentralized digital currency and payment system.
Scope	Broad technology with many possible applications.	Specific application of blockchain technology.
Purpose	Can be used for data sharing, smart contracts, supply chains, finance, etc.	Primarily designed for decentralized digital value transfer and payments.
Flexibility	Can be designed for many different applications and network models.	Its functionality is centered primarily around the Bitcoin protocol and monetary system.
Smart Contracts	Some blockchain platforms support advanced smart contracts.	Bitcoin supports programmable transaction functionality, but it is not a general-purpose smart-contract platform like Ethereum.
Examples	Ethereum, Hyperledger Fabric, Corda, etc.	Bitcoin itself.

Relationship

Bitcoin uses blockchain technology, but blockchain is not limited to Bitcoin.

PART B – 10 MARK QUESTIONS

Q1. Explain Blockchain Technology in Detail. Discuss Its Concept, Distributed Ledger Technology, Characteristics, and Importance.

Introduction

Blockchain is a distributed ledger technology that records transactions or other data in a sequence of cryptographically linked blocks. It enables multiple participants to maintain and verify a shared record without necessarily depending on one central authority.

Concept of Blockchain

The basic concept of blockchain involves:

1. **Transaction/Data:** Information is created by users.
2. **Verification:** Transactions are checked according to network rules.
3. **Consensus:** Participants agree on valid transactions.
4. **Block Formation:** Valid transactions are grouped into blocks.
5. **Cryptographic Linking:** Blocks are linked using hashes.
6. **Ledger Distribution:** Participating nodes maintain the accepted ledger state.

Basic Structure

Block 1 → Block 2 → Block 3 → Block 4

Each block generally contains:

- Transaction/data
- Timestamp
- Block hash
- Previous block hash
- Other network-specific information

Blockchain as Distributed Ledger Technology

A **Distributed Ledger Technology (DLT)** distributes a ledger among multiple participating computers or nodes.

Instead of:

Central Server → All Records

a distributed system can have:

Node A ↔ Node B ↔ Node C ↔ Node D

The participants maintain and validate the shared ledger according to the network's rules.

Characteristics of Blockchain

1. Decentralization

Control and ledger maintenance can be distributed across multiple participants.

2. Immutability

Confirmed records are designed to be highly resistant to unauthorized modification.

3. Transparency

Depending on the network type, participants can inspect and verify relevant records.

4. Security

Cryptography helps protect transactions and data.

5. Consensus

Participants use consensus mechanisms to agree on valid transactions and blockchain state.

6. Traceability

Records can provide an auditable history of transactions.

7. Programmability

Some blockchain platforms support smart contracts.

Importance of Blockchain

Blockchain can provide:

- Greater transparency.
- Tamper-resistant records.
- Shared data between organizations.
- Reduced dependence on intermediaries in suitable applications.
- Improved traceability.
- Automated business processes through smart contracts.
- A common record for multiple participants.

Applications

Blockchain can be used in:

- Cryptocurrency
- Banking and finance
- Supply-chain management
- Healthcare
- Digital identity
- Voting systems
- Asset/token management
- Smart contracts

Conclusion

Blockchain is an important form of distributed ledger technology that combines **cryptography, distributed networks, consensus, and linked data structures** to create a shared and verifiable record.

Q2. Explain the Different Types of Blockchain Networks in Detail. Compare Public, Private, Hybrid, and Consortium Blockchain Networks with Suitable Examples.

Blockchain networks can be classified according to **who can participate, who controls the network, and how access is managed**.

1. Public Blockchain

A public blockchain is generally open to anyone who follows the network's rules.

Features

- Open participation.
- High decentralization.

- Publicly verifiable transactions, depending on the design.
- No single organization necessarily controls the network.
- Uses a consensus mechanism.

Examples

- Bitcoin
- Ethereum

Applications

- Cryptocurrency
- Decentralized applications
- Public token networks

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2. Private Blockchain

A private blockchain is a **permissioned network** controlled by one organization or an authorized administrator.

Features

- Restricted access.
- Known participants.
- Centralized or organizational governance.
- Greater control over permissions.
- Suitable for enterprise use.

Example

- Hyperledger Fabric deployments

Applications

- Enterprise data sharing
- Supply-chain management
- Internal business processes

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3. Hybrid Blockchain

A **hybrid blockchain** combines characteristics of public and private blockchain systems.

Some information or activities can be kept restricted, while selected information can be made publicly accessible or verifiable.

Features

- Combination of permissioned and public characteristics.
- Organizations can control sensitive information.
- Selected information can be publicly verified.
- Provides flexibility between privacy and transparency.

Applications

- Healthcare
- Government services
- Supply chains
- Business systems requiring both privacy and public verification

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4. Consortium Blockchain

A **consortium blockchain** is a permissioned blockchain governed by a group of organizations rather than one organization.

Features

- Multiple organizations participate in governance.
- Access is restricted.
- Participants are generally known.
- Suitable for inter-organizational collaboration.

Examples

Enterprise blockchain networks based on platforms such as **Hyperledger Fabric** can be configured as consortium networks.

Applications

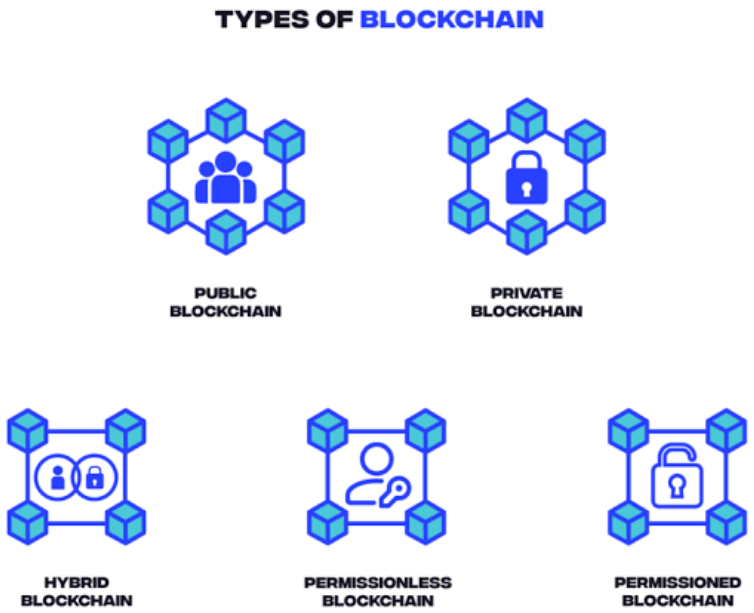
- Banking
- Supply-chain networks
- Insurance
- Trade and finance

Comparison

Feature	Public	Private	Hybrid	Consortium
Access	Generally open	Restricted	Partly restricted/open depending on design	Restricted
Control	Distributed	One organization	Combination	Multiple organizations
Participants	Anyone meeting network rules	Authorized users	Depends on design	Selected organizations
Transparency	Generally high	Controlled	Selective	Controlled among members
Decentralization	Generally high	Lower	Variable	Shared among participating organizations
Typical Use	Cryptocurrency, DApps	Enterprise	Mixed privacy/transparency needs	Inter-organizational business
Example	Bitcoin, Ethereum	Hyperledger Fabric deployment	Enterprise-specific hybrid systems	Consortium using Hyperledger Fabric

Conclusion

The choice of blockchain type depends on the required level of **access, privacy, decentralization, governance, and transparency**.



Q3. Differentiate Between Blockchain and Bitcoin in Detail Based on Definition, Scope, Purpose, Flexibility, Network Type, and Applications.

Introduction

Blockchain and Bitcoin are closely related but are **not the same thing**. Blockchain is a technology, while Bitcoin is a specific decentralized digital currency system that uses blockchain.

Detailed Comparison

Basis	Blockchain	Bitcoin
Definition	A distributed ledger technology for recording transactions or other data.	A decentralized digital currency and payment system.
Scope	Broad technology with many applications.	Specific application of blockchain technology.

Purpose	Designed to create shared, verifiable, and tamper-resistant records.	Designed for decentralized transfer of digital value without relying on a central monetary authority.
Flexibility	Can support different types of applications and network designs.	Primarily focused on the Bitcoin monetary and transaction system.
Network Type	Can be public, private, consortium, or hybrid depending on the implementation.	Public, decentralized blockchain network.
Participants	Depends on the blockchain design; can be open or permissioned.	Participants can join the public Bitcoin network according to its protocol.
Consensus	Can use different consensus mechanisms depending on the platform.	Uses Proof of Work (PoW).
Smart Contracts	Some blockchain platforms support general-purpose smart contracts.	Uses Bitcoin's scripting system for transaction conditions; it is not a general-purpose smart-contract platform like Ethereum.
Applications	Finance, supply chains, healthcare, identity, asset management, DApps, etc.	Digital currency, payments, and transfer/store of digital value.
Examples	Ethereum, Hyperledger Fabric, Corda, Quorum, etc.	Bitcoin.
Relationship	Blockchain can be used to build systems such as Bitcoin.	Bitcoin is an application/system built using blockchain technology.

Key Relationship

The easiest way to understand the relationship is:

Blockchain = Technology
Bitcoin = A digital currency system that uses blockchain

For example, just as the **Internet** supports many applications such as email and websites, **blockchain technology** can support many applications, with Bitcoin being one important example.

Applications of Blockchain

Blockchain technology can be applied to:

1. Banking and finance
2. Supply-chain tracking
3. Healthcare records
4. Digital identity
5. Smart contracts
6. Tokenization
7. Decentralized applications

Applications of Bitcoin

Bitcoin is mainly associated with:

1. Digital payments
2. Peer-to-peer transfer of value
3. Holding and transferring bitcoin as a digital asset

Conclusion

Blockchain is the underlying technology, whereas Bitcoin is a specific system and cryptocurrency that uses blockchain. Blockchain has a much broader scope and can be adapted to many applications and network models, while Bitcoin has a specific monetary purpose and protocol.

PART A – 5-MARK QUESTIONS

Q1. What is a Blockchain Wallet? Explain its Key Components.

A **blockchain wallet** is a software, hardware, or other mechanism that allows a user to **manage blockchain assets and interact with blockchain networks**. It does not literally store coins; instead, it manages the **cryptographic keys** used to control assets recorded on the blockchain.

Key Components

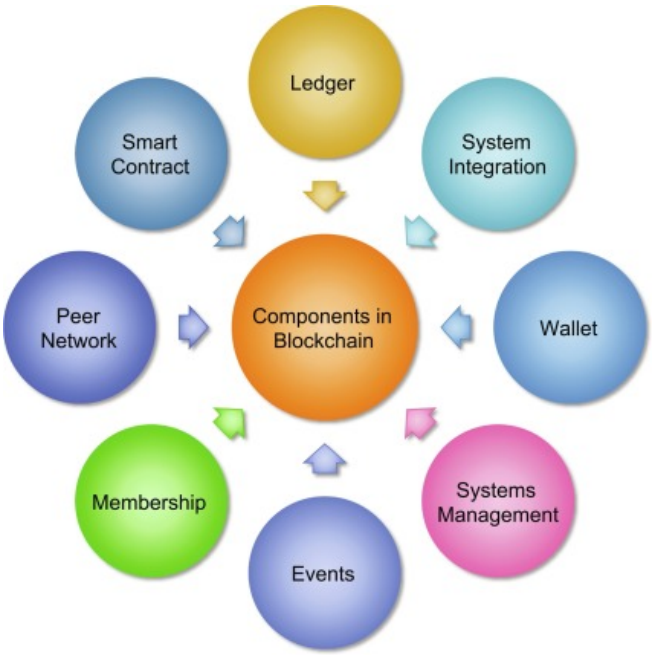
1. **Public Address**
   A blockchain address is used to receive assets. It can generally be shared with others.
2. **Private Key**
   A secret cryptographic key used to authorize transactions. Anyone who controls it may be able to control the associated assets.

3. **Seed Phrase**
   A human-readable sequence of words that can be used to recover a wallet and its keys.
4. **Wallet Software/Interface**
   Allows users to view balances, create transactions, and interact with blockchain applications.
5. **Key Management System**
   Generates, stores, and protects cryptographic keys.

Example

A user can use a wallet such as **MetaMask** to manage Ethereum-based assets and interact with decentralized applications.

Conclusion: A wallet primarily manages the keys and information required to access and control blockchain assets.



Q2. Differentiate Between Hot Wallets and Cold Wallets with Examples.

Hot Wallet	Cold Wallet
Connected to the Internet or designed for frequent online use.	Keeps keys offline when not in use.
Convenient for frequent transactions.	Better suited to long-term key storage.

More exposed to online threats.	Reduces exposure to online attacks.
Easy and quick to access.	Usually less convenient for frequent transactions.
Examples: MetaMask, Trust Wallet	Examples: Ledger, Trezor

Hot Wallet

A **hot wallet** is connected to the Internet or operates in an environment with online connectivity.

Example: MetaMask browser/mobile wallet.

Cold Wallet

A **cold wallet** keeps private keys offline, reducing their exposure to Internet-based attacks.

Example: Ledger hardware wallet.

Conclusion: Hot wallets provide convenience, while cold wallets are generally preferred for reducing online exposure when storing assets for longer periods.

Q3. Define a Smart Contract. List Its Key Characteristics.

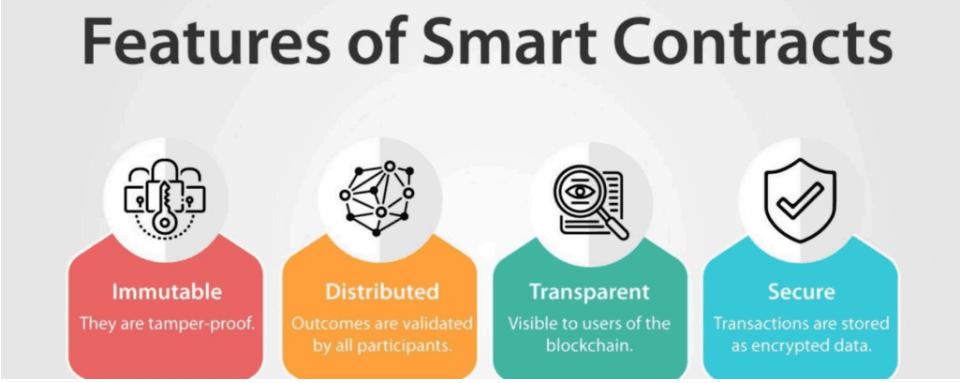
Definition

A **smart contract** is a program deployed on a blockchain that automatically executes predefined rules when specified conditions are satisfied.

Key Characteristics

1. **Self-executing** – Executes according to programmed conditions.
2. **Automated** – Reduces the need for manual processing.
3. **Transparent** – Contract code and execution can be verifiable depending on the blockchain.
4. **Tamper-resistant** – Deployed contract code is difficult to change unless upgrade mechanisms are built in.
5. **Deterministic** – Given the same valid inputs and blockchain state, execution follows defined rules.
6. **Programmable** – Can implement business rules and transactions.
7. **Reduced Intermediaries** – Can automate processes that might otherwise require intermediaries.

Example: A smart contract can automatically transfer a digital asset when predefined payment conditions are fulfilled.



Q4. Explain How Proof of Work (PoW) Reaches Consensus. State Its Advantages and Limitations.

Meaning

Proof of Work (PoW) is a consensus mechanism in which participants called **miners** compete to solve a computational puzzle.

Working

1. Transactions are collected into a candidate block.
2. Miners repeatedly change a value called a **nonce**.
3. They calculate the block's cryptographic hash.
4. The goal is to find a hash satisfying the network's difficulty requirement.
5. The miner that finds a valid solution broadcasts the block.
6. Other nodes verify the solution and block.
7. The accepted block becomes part of the blockchain according to the protocol's rules.

Advantages

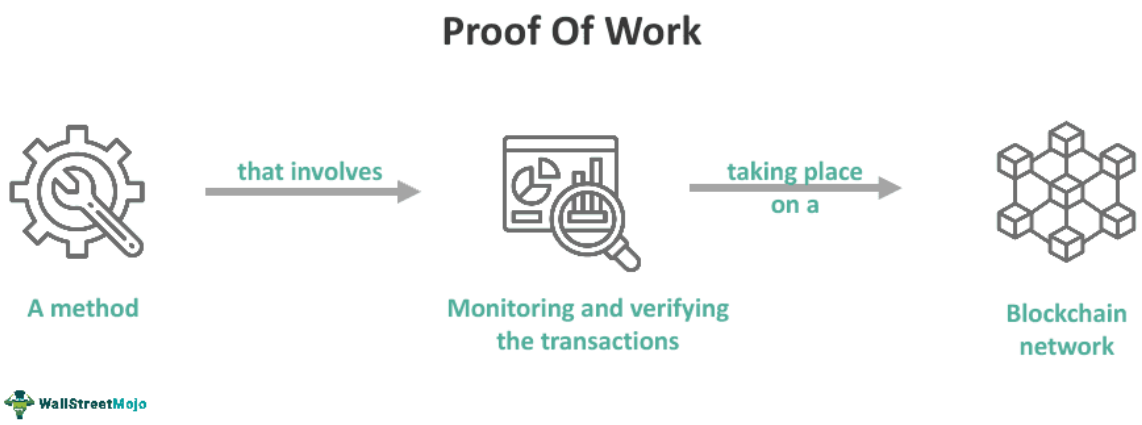
- Strong Sybil resistance through computational work.
- Well-established security model.
- Open participation in suitable public networks.
- Can make rewriting historical records computationally expensive.

Limitations

- High energy consumption.
- Requires specialized hardware in some networks.
- Transaction confirmation can be relatively slow.

- Mining can become concentrated among large operators.

Example: Bitcoin uses Proof of Work.



Q5. Explain How Proof of Stake (PoS) Works. How Does "Slashing" Discourage Dishonest Behaviour?

Meaning

Proof of Stake (PoS) is a consensus mechanism in which validators participate by locking or **staking** cryptocurrency as economic collateral.

Working

1. Users lock the required cryptocurrency as stake.
2. The protocol selects validators according to its rules.
3. Selected validators propose or attest to blocks.
4. Other validators verify the proposed blocks.
5. Valid blocks are accepted by the network.
6. Honest validators can receive rewards according to the protocol.

Slashing

Slashing is a penalty mechanism used in some PoS systems.

If a validator performs prohibited actions, such as certain forms of conflicting or malicious validation, the protocol may destroy or reduce part of its staked assets.

How Slashing Discourages Dishonesty

Potential dishonest behaviour → Risk of losing stake → Economic incentive to behave honestly

Advantages

- Much lower energy consumption than PoW.
- Can support efficient transaction processing.
- Economic incentives encourage correct participation.

Example: Ethereum currently uses Proof of Stake.

Q6. What is Proof of Burn (PoB)? Explain How It Grants Mining/Validation Rights.

Proof of Burn (PoB) is a consensus approach in which participants permanently **destroy cryptocurrency tokens** to demonstrate commitment to the network.

Working

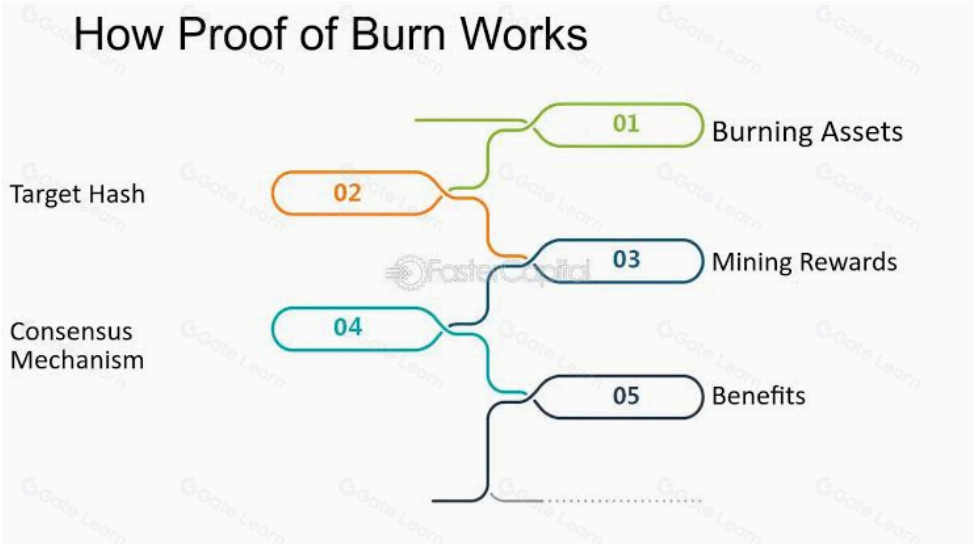
1. A participant sends coins to an address from which they cannot be recovered.
2. The coins are considered **burned**.
3. The participant gains eligibility or increased influence in the consensus process according to the particular PoB protocol.
4. Participants with sufficient burn commitment can obtain mining/validation opportunities.
5. Honest participation can result in rewards according to the network's rules.

Features

- Does not require the same computational mining as PoW.
- Creates an economic cost through permanently destroyed tokens.
- Can provide Sybil resistance.
- Reduces the circulating supply of the burned asset.

Limitation

The permanent destruction of tokens can be economically costly.



Q7. Explain the Working of Raft Consensus, Including the Roles of Leader, Follower, and Candidate.

Raft is a consensus algorithm designed for **distributed systems** where a group of nodes needs to maintain a consistent replicated log. It is commonly used in permissioned/distributed systems rather than open cryptocurrency networks.

Roles

1. Leader

- Handles client requests.
- Replicates log entries to followers.
- Sends periodic heartbeats.
- Coordinates the replicated log.

2. Follower

- Receives messages from the leader.
- Replicates log entries.
- Does not normally initiate operations.
- Can become a candidate if it stops receiving heartbeats.

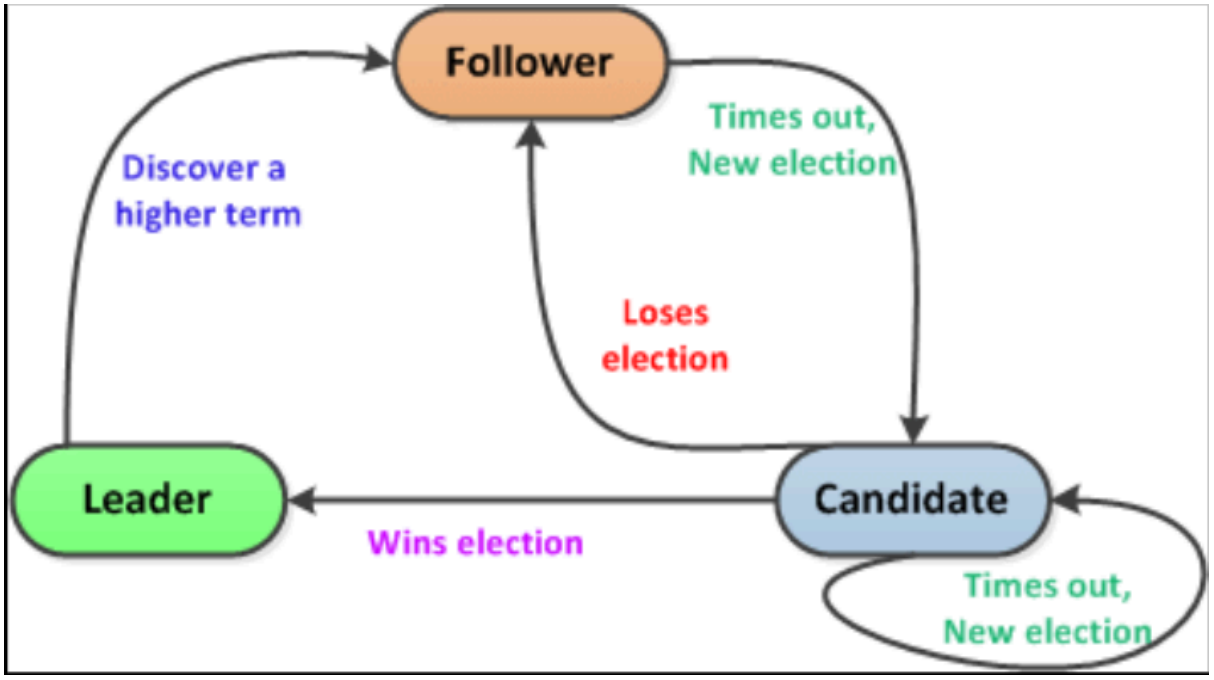
3. Candidate

- A follower becomes a candidate when an election timeout occurs.
- Requests votes from other nodes.
- If it receives the required majority, it becomes the leader.

Basic Flow

Follower → Candidate → Leader

The leader replicates entries to followers and an entry is considered committed when the required majority has accepted it according to Raft's rules.



Q8. Explain Proof of Capacity (PoC) and How It Uses Storage Instead of Computation.

Proof of Capacity (PoC) is a consensus mechanism that uses available **storage space** rather than continuous computational work as the main resource for participation.

Working

1. A participant allocates storage space.
2. Data or cryptographic values are generated and stored on the disk.
3. These stored values are sometimes called **plots**.
4. When a new block needs to be created, the participant checks its stored data.
5. The protocol determines which participant has the appropriate proof according to its rules.
6. The selected participant can create the block and receive a reward.

Advantages

- Lower computational requirements than PoW.
- Can consume less energy during the consensus process.
- Uses existing storage resources.

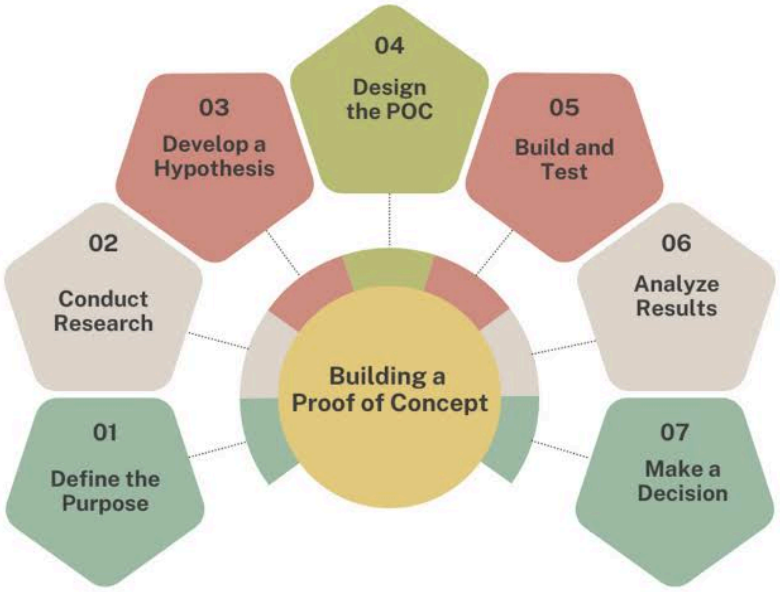
Limitation

Large amounts of storage may be required, and specialized storage hardware can create environmental and economic concerns.

Q9. Differentiate Between Proof of Work and Proof of Stake Based on Energy Consumption, Speed, and Basis of Consensus.

Basis	Proof of Work (PoW)	Proof of Stake (PoS)
Consensus basis	Computational work	Economic stake
Participants	Miners	Validators
Energy consumption	Generally high	Generally much lower
Block creation	Based on solving computational puzzles	Validators are selected according to protocol rules
Speed	Can have comparatively slower confirmation depending on network design	Can provide faster/efficient confirmation depending on protocol
Hardware requirement	Can require significant computing resources	Does not require PoW-style mining hardware
Security incentive	Cost of computation/electricity	Economic value at risk through staking
Example	Bitcoin	Ethereum

Conclusion: PoW secures the network through computational expenditure, while PoS uses economic stake and validator incentives.



PART B – 10-MARK QUESTIONS

Q1. Explain in Detail the Various Types of Wallets Based on Connectivity, Custody, and Form Factor, with Examples.

A **blockchain wallet** is a tool for managing cryptographic keys and interacting with blockchain networks. Blockchain assets themselves remain recorded on the blockchain; the wallet manages the keys that authorize transactions.

Wallets can be classified in three major ways:

A. Based on Connectivity

1. Hot Wallet

A hot wallet is connected to the Internet or designed for online use.

Features:

- Easy access.
- Convenient for frequent transactions.
- Higher exposure to online threats.

Examples: MetaMask, Trust Wallet.

2. Cold Wallet

A cold wallet keeps private keys offline when not being used.

Features:

- Reduced online exposure.
- Suitable for long-term storage.
- Less convenient for frequent transactions.

Examples: Ledger, Trezor hardware wallets.

B. Based on Custody

1. Custodial Wallet

In a custodial wallet, a third party or service provider controls the private keys on behalf of the user.

Example: A cryptocurrency exchange account where the exchange holds the keys.

Advantages:

- Easy for beginners.
- Account recovery may be available.
- Convenient for trading.

Disadvantage: The user depends on the custodian to safeguard and provide access to the assets.

2. Non-Custodial Wallet

In a non-custodial wallet, the user controls the private keys or recovery phrase.

Examples: MetaMask and many hardware wallets.

Advantages:

- User has direct control.
- No dependency on an exchange for key custody.

Disadvantage: Losing the recovery phrase or private key can result in loss of access.

C. Based on Form Factor

1. Software Wallet

Runs as a mobile application, desktop program, or browser extension.

Examples: MetaMask, Trust Wallet.

2. Hardware Wallet

A dedicated physical device designed to protect private keys.

Examples: Ledger, Trezor.

3. Paper Wallet

A historical approach where keys or addresses were printed or written on paper.

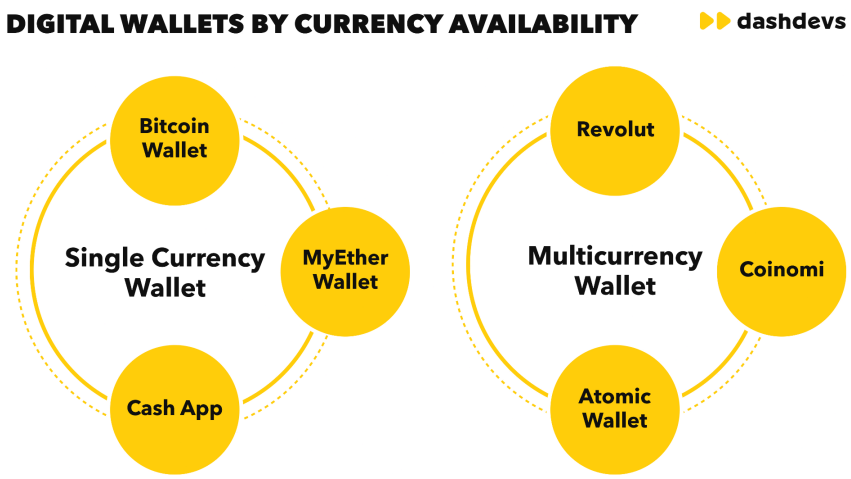
Limitation: It can be lost, damaged, copied, or destroyed and is generally less convenient than modern wallet solutions.

Summary

Classification	Types	Examples
Connectivity	Hot, Cold	MetaMask, Ledger
Custody	Custodial, Non-custodial	Exchange wallet, MetaMask
Form factor	Software, Hardware, Paper	Trust Wallet, Trezor, printed keys

Conclusion

Wallet classification helps users understand **how keys are connected, who controls them, and how they are stored**. Security, convenience, recovery, and intended usage should all be considered when choosing a wallet.



Q2. Define a Smart Contract and Explain Its Key Characteristics and Advantages with Real-World Examples.

Definition

A **smart contract** is a program deployed on a blockchain that automatically executes predefined rules when specified conditions are met.

For example:

Condition → Verification → Automatic Execution

Key Characteristics

1. Automation

Smart contracts can execute programmed actions automatically.

2. Transparency

Depending on the blockchain, contract code and transactions can be inspected by participants.

3. Deterministic Execution

The same valid inputs and blockchain state produce the defined result according to the contract's code.

4. Tamper Resistance

Deployed contract code is generally difficult to modify unless the contract has been designed with an upgrade mechanism.

5. Programmability

Developers can create applications with complex business logic.

6. Security Through Cryptography

Blockchain accounts and transactions use cryptographic mechanisms for authorization.

7. Reduced Intermediaries

Some processes can be automated without requiring a traditional intermediary for every step.

Advantages

1. **Automation** – Reduces manual processing.
2. **Efficiency** – Can execute predefined operations quickly.
3. **Transparency** – Relevant transaction history can be independently verified.
4. **Accuracy** – Code executes according to defined rules.
5. **Traceability** – Blockchain records provide an auditable history.
6. **Reduced Intermediary Dependence** – Suitable processes can be automated.

Real-World Examples

1. Supply Chain

A smart contract can record delivery events and release payment when predefined delivery conditions are met.

2. Insurance

A contract could automatically process a claim when trusted data satisfies predefined conditions.

3. Decentralized Finance

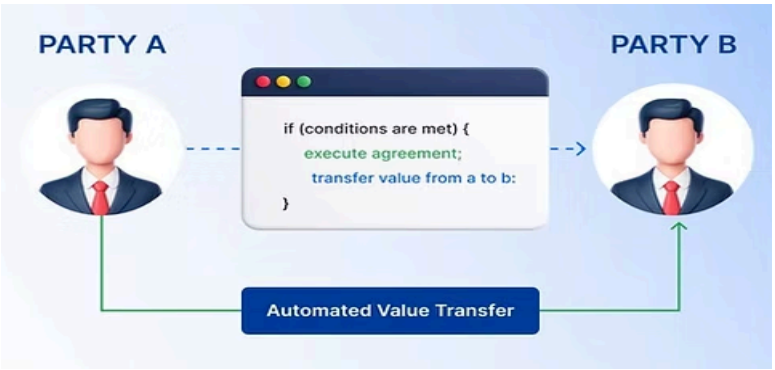
Smart contracts can automate activities such as token exchanges, lending, and borrowing according to protocol rules.

4. Digital Assets

Smart contracts can define ownership and transfer rules for blockchain-based tokens.

Conclusion

Smart contracts combine **programming, cryptography, and blockchain** to automate predefined processes. They are widely used in decentralized applications and can also support enterprise processes.



Q3. Explain the Working, Key Components, and Functions of a Blockchain Wallet, and Justify Why a Wallet Does Not Literally "Hold" Coins.

Introduction

A blockchain wallet is a software or hardware tool used to **manage cryptographic keys and interact with blockchain networks**.

A common misconception is that coins are physically stored inside the wallet.

Important Concept

Coins/tokens are recorded on the blockchain.

The wallet stores or manages the **keys** that allow the user to authorize transactions involving those assets.

Key Components

1. Public Address

The address is used to receive assets.

It can generally be shared with other people.

2. Private Key

The private key is secret and is used to authorize transactions.

Private key = Control/authorization

It must be protected carefully.

3. Seed/Recovery Phrase

A seed phrase is a sequence of words used by many wallets to derive or recover cryptographic keys.

4. Wallet Application

The application provides the user interface for:

- Viewing balances
- Sending assets
- Receiving assets
- Connecting to decentralized applications
- Signing transactions

5. Network Information

The wallet interacts with a particular blockchain network and obtains information such as account state and transaction status.

Working of a Wallet

Step 1: Key Generation

The wallet generates or imports cryptographic keys.

Step 2: Address Generation

A public blockchain address is derived according to the blockchain's rules.

Step 3: Receiving Assets

Another user sends assets to the wallet's blockchain address.

Step 4: Blockchain Records the Transaction

The transaction is validated and recorded on the blockchain.

Step 5: Sending Assets

The wallet creates a transaction and signs it using the appropriate private key.

Step 6: Network Verification

The blockchain network verifies the transaction according to its protocol.

Step 7: Ledger Update

After confirmation, the blockchain records the new state.

Why Doesn't a Wallet Literally Hold Coins?

Suppose a user has **1 ETH**.

The ETH is not stored inside the phone or hardware wallet like a file.

Instead:

Blockchain → Records the asset/account state

Wallet → Manages the keys needed to authorize transactions

Therefore, if the wallet device is lost but the recovery credentials are securely backed up, the user may be able to recover access to the assets through another compatible wallet.

Conclusion

A blockchain wallet is best understood as a **key-management and blockchain-interaction tool**, not as a physical container for cryptocurrency.



Q4. Explain Proof of Activity (PoA) in Detail. Describe How It Combines Proof of Work and Proof of Stake and Discuss Its Advantages and Limitations.

Meaning

Proof of Activity (PoA) is a hybrid consensus approach that combines concepts from **Proof of Work (PoW)** and **Proof of Stake (PoS)**.

Its basic idea is to use computational work to initiate block creation and then use stakeholders to participate in completing/validating the block.

> **Note:** Here, PoA means **Proof of Activity**, not Proof of Authority.

Working of Proof of Activity

Step 1: Mining Starts

Miners perform computational work similar to PoW.

Step 2: Block Template

A miner finds a valid PoW solution and creates a block containing information that allows the next stage of validation.

Step 3: Stakeholder Selection

The protocol selects a group of stakeholders/validators based on its rules.

Step 4: Validation

Selected stakeholders participate in validating or signing the block.

Step 5: Block Completion

Once the required participation is obtained, the block is accepted according to the protocol's rules.

Step 6: Rewards

Mining and validation participants receive rewards according to the specific implementation.

Simple Flow

PoW Mining → Stakeholder Selection → Validation → Block Confirmation → Rewards

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Advantages

1. Combination of Two Security Models

It combines computational work with economic stake.

2. Reduced Dependence on Continuous PoW

The hybrid design can reduce some of the computational burden compared with a pure PoW system, depending on implementation.

3. Economic Incentives

Stakeholders have economic incentives to participate honestly.

4. Additional Security Layer

An attacker may need to overcome both computational and staking-related requirements.

Limitations

1. Complexity

The system is more complicated than using a single consensus mechanism.

2. Energy Consumption

It still involves PoW, so it can consume significant energy.

3. Stake Concentration

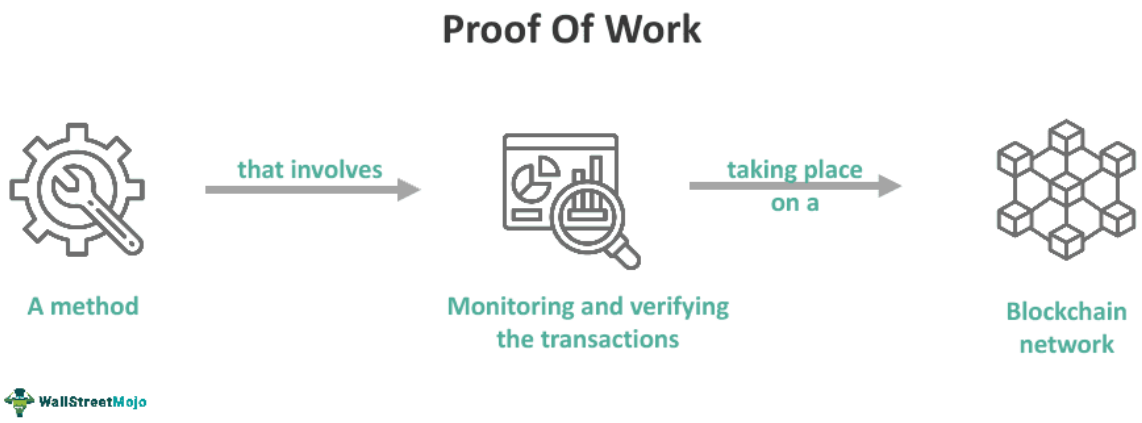
Large stakeholders may have greater influence depending on the protocol.

4. Implementation Dependence

Security and efficiency depend heavily on the specific design of the PoA system.

Conclusion

Proof of Activity combines **computational work and stake-based participation** to create a hybrid consensus mechanism, attempting to obtain useful properties from both approaches.



Q5. Compare Proof of Work, Proof of Stake, Proof of Burn, Raft, Federated Consensus, Proof of Activity, PoET, Proof of Capacity, and Proof of Importance Based on Energy Use, Speed, Decentralization, Sybil Resistance, and Consensus Basis.

Comparison Table

Consensus	Energy Use	Speed	Decentralization	Sybil Resistance	Consensus Basis
Proof of Work (PoW)	High	Relatively slower	Generally high in open networks	Computational cost	Computing power/work

Proof of Stake (PoS)	Low	Generally faster than PoW, depending on protocol	Generally high in suitable public networks	Economic stake	Staked assets
Proof of Burn (PoB)	Low–moderate	Generally faster than PoW, depending on design	Variable	Economic cost through burning	Destroyed/burned tokens
Raft	Very low	Fast	Low-to-moderate; permissioned	Membership/controlled participation	Majority agreement among known nodes
Federated Consensus	Low	Fast	Lower than open public systems; relies on selected validators	Trusted/controlled validator membership	Agreement among selected nodes
Proof of Activity (PoA)	Moderate–high	Moderate	Variable	Combination of work and stake	PoW + PoS-style participation
Proof of Elapsed Time (PoET)	Low	Generally fast	Depends on validator model	Trusted execution/lottery mechanism	Random wait time, typically via trusted hardware
Proof of Capacity (PoC)	Lower computational energy than PoW	Moderate	Variable	Allocated storage capacity	Available storage space
Proof of Importance (PoI)	Low	Generally faster than PoW	Variable	Economic/network participation factors	Importance score

Explanation of the Major Mechanisms

1. Proof of Work

Participants compete using computational power. The winning miner proposes a block.

Main resource: Computing power.

2. Proof of Stake

Validators lock assets as stake and participate according to protocol rules.

Main resource: Economic stake.

3. Proof of Burn

Participants permanently destroy tokens to demonstrate economic commitment.

Main resource: Burned cryptocurrency.

4. Raft

Known nodes elect a leader and replicate logs through majority agreement.

Main resource: Trusted/permissioned node membership.

5. Federated Consensus

A predefined or selected group of trusted validators reaches agreement.

Main resource: Membership and validator trust.

6. Proof of Activity

Combines PoW-style mining with PoS-style validation.

Main resources: Computation + stake.

7. Proof of Elapsed Time (PoET)

Participants wait for randomly assigned periods, typically using trusted execution environments, and the participant whose wait ends according to protocol rules may propose the block.

Main resource: Trusted waiting/lottery mechanism.

8. Proof of Capacity

Participants use available disk storage as the resource for consensus.

Main resource: Storage capacity.

9. Proof of Importance

Participants are selected based on an importance score that can consider factors such as stake and network activity, depending on the protocol.

Main resource: Network/economic importance.

Overall Comparison

Energy Consumption

PoW generally has the highest energy requirement because it relies on continuous computational work. PoS, PoB, Raft, Federated Consensus, and PoET generally require much less computational energy.

Speed

Permissioned approaches such as **Raft** and federated consensus can provide fast agreement because they operate among known participants. Public consensus mechanisms may make different trade-offs between decentralization, security, and speed.

Decentralization

Public PoW and PoS networks can support broad participation, while Raft and federated systems normally depend on a controlled set of participants.

Sybil Resistance

- **PoW:** computational expenditure
- **PoS:** economic stake
- **PoB:** economic cost through burning
- **Raft/Federated:** controlled membership
- **PoC:** storage commitment
- **PoET:** trusted execution/lottery mechanism
- **PoI:** economic/network activity-based importance

Conclusion

There is no single consensus mechanism that is ideal for every blockchain. The appropriate mechanism depends on the required balance of **security, energy efficiency, transaction speed, decentralization, governance, and participation model**.